GCA Session 5 - Lab Report

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Abstract: This project presents an analysis of synchronous and asynchronous execution on the GPU by implementing synchronous and asynchronous versions of reduction kernels for finding maximum, minimum, sum, and product of elements in an array. Results demonstrate notable differences in execution time and efficiency between the two approaches.
Keywords: CUDA, GPU, CPU, C++

1 Concepts

This report discusses the difference in execution time for synchronous and asynchronous programs on the GPU.

1.1 Synchronous

Synchronous execution in GPU programming means launching a kernel and doing all memory transfers sequentially. Each operation will wait for the previous operation to finish before proceeding. This is the most easy to understand way of programming because it is so predictable. However, it also misses out on some potential benefits that asynchronous programming tries to exploit. For instance, cores may remain idle during synchronization.

1.2 Asynchronous

Asynchronous execution might be a bit more difficult to comprehend, as it differs a lot from the traditional synchronous approach. In the context of the project that was build to detect the maximum, minimum, sum, and product of an array of elements, asynchronous execution was employed to achieve parallelism in memory transfers. By utilizing CUDA streams, asynchronous execution facilitated overlapping of data transfers and computation. This approach resulted in efficient use of the GPU and had shorter overall computation times compared to synchronous execution.

2 Project

To demonstrate the concepts discussed earlier, I developed a project that detected the maximum, minimum, sum and product of elements in an array. This happened both synchronously and asynchronously and was executed 1 000 times to get the results. The graph below shows the total execution time in milliseconds of these 1 000 runs. asynchronous execution is on the left, synchronous is on the right.µ

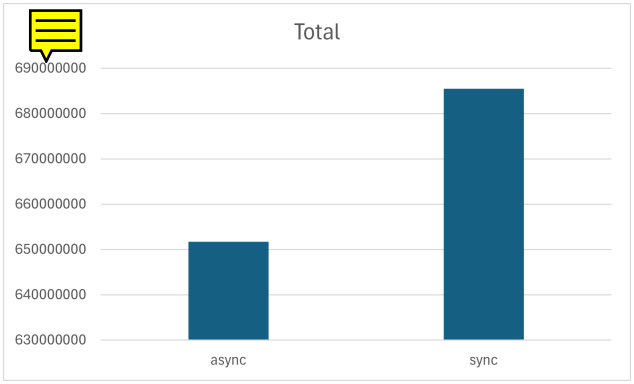


Figure 1: *illustration of execution time for asynchronous vs synchronous*

3 Github link

`https://github.com/StijnVandendriessche1/GC_lab5.git`